

GENE EXPRESSION CONTROL AND REGULATIONS

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Objectives:

At the end of this chapter, it is expected;

- To describe the steps involved in the processes of gene expression (both in prokaryotes and eukaryotes)**
- To appreciate that DNA-binding transcription factors, proteins that bind to specific DNA sequences that are physically linked to their target transcriptional promoter elements, can either activate or repress gene transcription.**
- To recognize that DNA-binding transcription factors are often modular proteins that are composed of structurally and functionally distinct domains, which can directly or indirectly control transcription of messenger RNA (mRNA).**
- To recognize the special features in the gene unit, which involve in control of gene expression**

IMPORTANCES OF GENE EXPRESSION CONTROL AND REGULATION

- **Organisms adapt to environmental changes by altering gene expression that are necessary for the;**
 - growth,**
 - development,**
 - differentiation and**
 - the very existence of the organism.**

CONTROL OF GENE EXPRESSION

- ◉ **In prokaryotes:**
 - ◉ **Studied in detail**
 - ◉ **Involves the interaction of specific binding proteins with various regions of DNA;**
 - **In the immediate vicinity of the transcription site**
 - **Produces either a positive or negative effect on transcription.**
- ◉ **In eukaryotes:**
 - ◉ **Studies are not extensive and not well understood.**
 - **Cells in eukaryotes utilize the same basic principle but uses other mechanisms to regulate transcription.**

Types of Gene Expression Regulation

Two types:

- 1. Positive regulation**
- 2. Negative regulation**

Positive regulation:

- **Positive regulation:**

- When the expression of genetic information is quantitatively increased by the presence of specific regulatory element.

- **Positive regulator:**

- The element or molecule mediating positive regulation.

Negative regulation:

- **Negative regulation:**

- **When the expression of genetic information is decreased by the presence of a specific regulatory element.**

- **A double negative has the effect of acting as a positive.**
 - **An effector that inhibits the function of a negative regulatory appears to bring about a positive regulation.**

- **Negative regulator:**

- **The element or molecule mediating the negative regulation.**

One Cistron-One Subunit Concept

- **Earlier concept: “one gene-one enzyme” hypothesis.**
 - **Beadle and Tatum, 1941.**
 - **Some enzymes and protein molecules are composed of two or more non-identical subunits, which cannot be explained by “one gene-one enzyme” theory and so it is not valid.**
- **Cistron and One cistron-one subunit concept:**
 - **The genetic unit coding for the subunit of an enzyme or protein molecule.**
 - **The smallest unit of genetic expression.**
 - **Hence, the “one gene-one enzyme” idea might be more accurately regarded as “One cistron-one polypeptide concept”.**

Type of genes:

- **Inducible genes:**

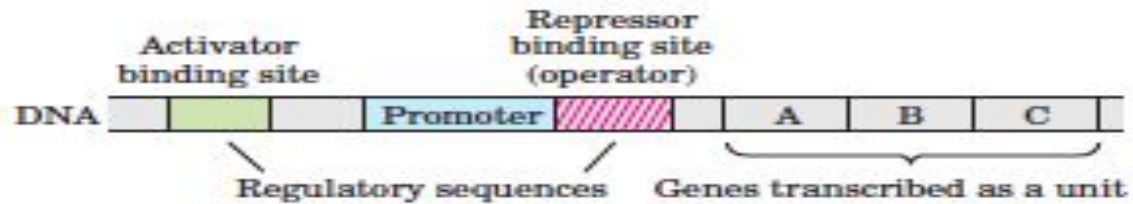
- **The expression of the inducible gene increases in response to an inducer.**
- **Inducers are small molecules.**

- **Constitutive genes (housekeeping genes):**

- **Expressed at more or less constant rate in almost all the cells**
- **They are not subjected to regulation.**
- **The products of these genes are required all the time in a cell.**

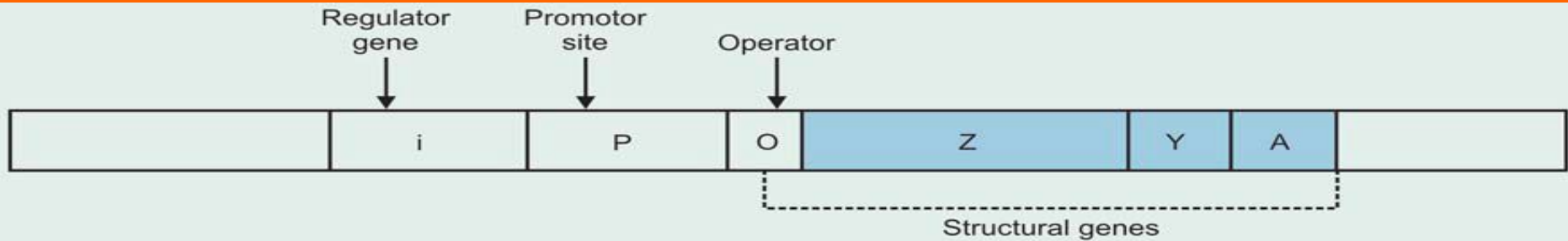
Control of Gene Expression in Prokaryotes

Operon Model:



- Concept introduced by Jacob and Monod in 1961.
- Operon is defined as a segment of a DNA strand consisting of:
 - Structural genes
 - Code for proteins & RNAs except regulatory factors. Appears as cluster.
 - Operator gene
 - Has an overall control over the process of translation
 - Regulator gene
 - Responsible for synthesis of repressor protein molecules which regulate the transcription
 - P site (promoter site)
 - Situated between operator gene and regulator gene

“Lac-operon”:



- ❖ The “lac operon” is an inducible catabolic operon of *E. coli*.
- ❖ It consists of
 - ❑ Structural genes: ‘Z’, ‘Y’ and ‘A’
 - ❑ Code respectively for β -galactosidase, galactoside permease and thiogalactoside transacetylase.
 - ❑ Operator gene or “O” gene:
 - ❑ Located immediately on the 3' or upstream side of the ‘Z’ gene
 - ❑ A regulator or “i” gene:
 - ❑ Located on the same DNA strand on the upstream side of the ‘O’ gene and transcribes an m-RNA which codes for repressor proteins (a tetramer).
 - ❑ Promoter site (P site):
 - ❑ Situated next of the operator gene where the enzyme RNA polymerase binds and it makes the structural genes Z, Y and A to transcript.

Functions of Lactose metabolizing Enzymes:

❖ β -galactosidase:

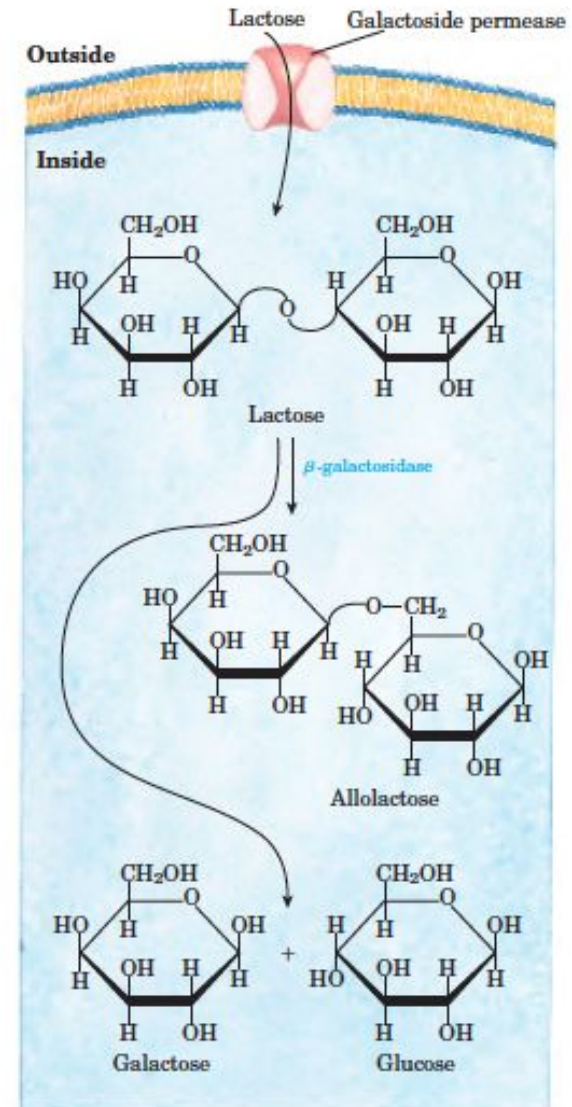
- ❑ Coded by 'Z' gene.
- ❑ Hydrolyses lactose (β -galactoside) to galactose and glucose.

❖ Galactoside Permease:

- ❑ Coded by 'Y' gene.
- ❑ Responsible for the transport of lactose into the cell.

❖ Thiogalactoside transacetylase:

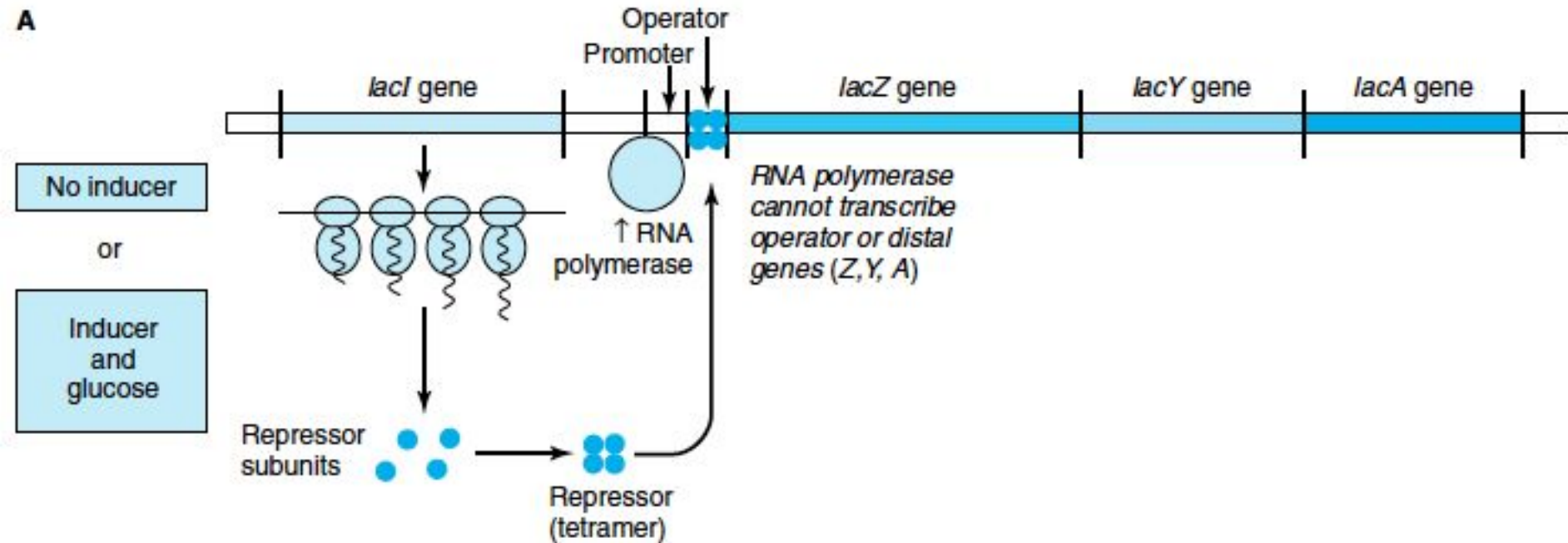
- ❑ Coded by 'A' gene
- ❑ Function is not known properly.
- ❑ However, the enzyme's cellular role may be to modify toxic galactosides to facilitate their removal from the cell.



Repression of 'Lac' Operon:

- **Regulator gene (i gene):**
 - A constitutive gene - is independent of inducer.
 - Expressed at a constant rate leading to synthesis of subunits of the 'Lac' repressors □ These subunits assemble into a 'Lac' repressor molecule
- **Lac' repressor:**
 - A tetramer protein - each monomer has a molecular weight of 40,000.
 - This repressor protein molecule has a high affinity for "operator" gene (O) and it specifically binds to the operator gene (O).
- **Repression of 'Lac' operon:**
 - The binding of repressors with 'O' gene prevents the binding of RNA polymerase to the promoter site (P) □ thereby preventing the transcription of structural genes Z, Y and A which are inducible enzymes.
 - This is repression of 'Lac' operon which occurs in E. coli in absence of lactose and transcription of structural genes do not occur.
 - The repressor molecules act as negative regulator of gene expression.

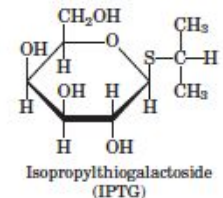
Mechanism of Repression of 'Lac' operon:



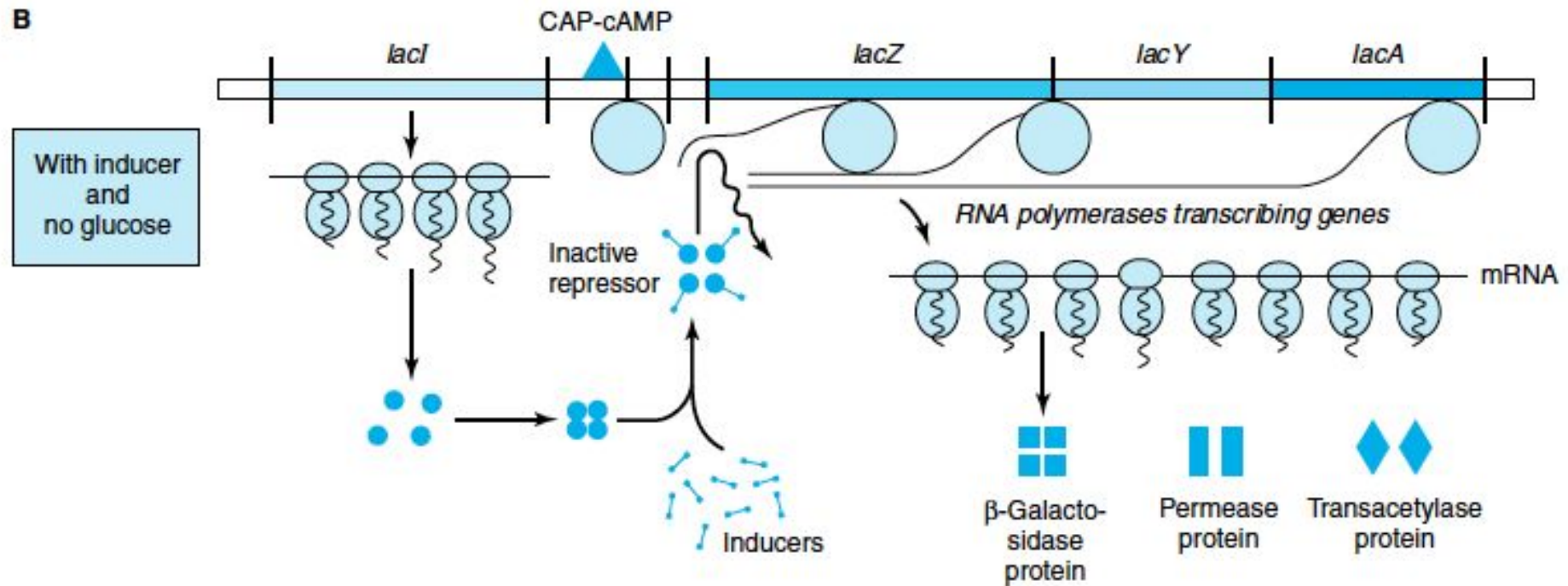
- ◉ When either no inducer is present or inducer is present with glucose;
 - The lacI gene products that are synthesized constitutively form a repressor tetramer molecule
 - The repressor binds at the operator locus to prevent the efficient initiation of transcription by RNA polymerase at the promoter locus
 - Thus prevents the subsequent transcription of the lacZ, lacY, and lacA structural genes.

Derepression of 'Lac' Operon:

- **Derepression/ Induction of 'Lac' operon:**
 - When there is addition of inducer or a gratuitous inducer results in prompt derepression and induction.
 - Inducer: Lactose
 - Gratuitous inducer: Isopropyl thiogalactoside (IPTG), a structural analogue of lactose
 - The inducer molecules react with the repressor molecules attached to the operator locus as well as in cytoplasm
 - Brings about inactivation of repressor molecules by changing the conformation of the molecules.
 - The DNA-dependent RNA polymerase attaches to the DNA at the promoter site (P) and permeates to operator locus which is now free;
 - Transcription of structural genes takes place leading to formation of polycystronic mRNA and finally the three enzymes.



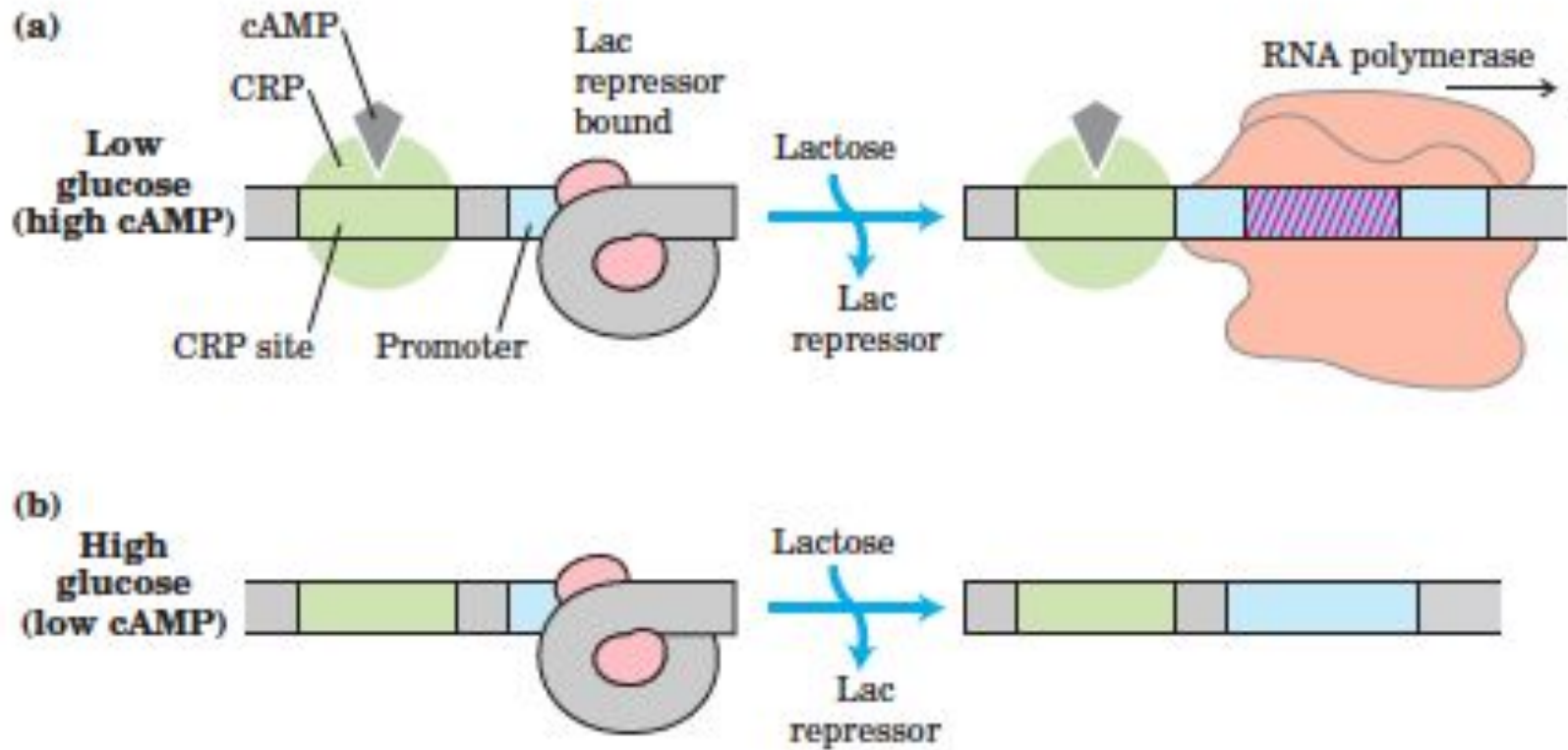
Mechanism of Derepression of 'Lac' operon:



- When inducer is present;
 - The constitutively expressed *lacI* gene forms repressor molecules that are conformationally altered by the inducer
 - Altered repressor cannot efficiently bind to the operator locus (affinity of binding reduced > 1000-fold).
 - In the presence of cAMP and its binding protein (CAP), the RNA polymerase can transcribe the structural genes *lacZ*, *lacY*, and *lacA*, and the polycistronic mRNA molecule formed.
 - mRNA can be translated into the corresponding protein molecules β -galactosidase, permease, and transacetylase, allowing for the catabolism of lactose.

Role of CAP-cAMP complex - Positive Regulation

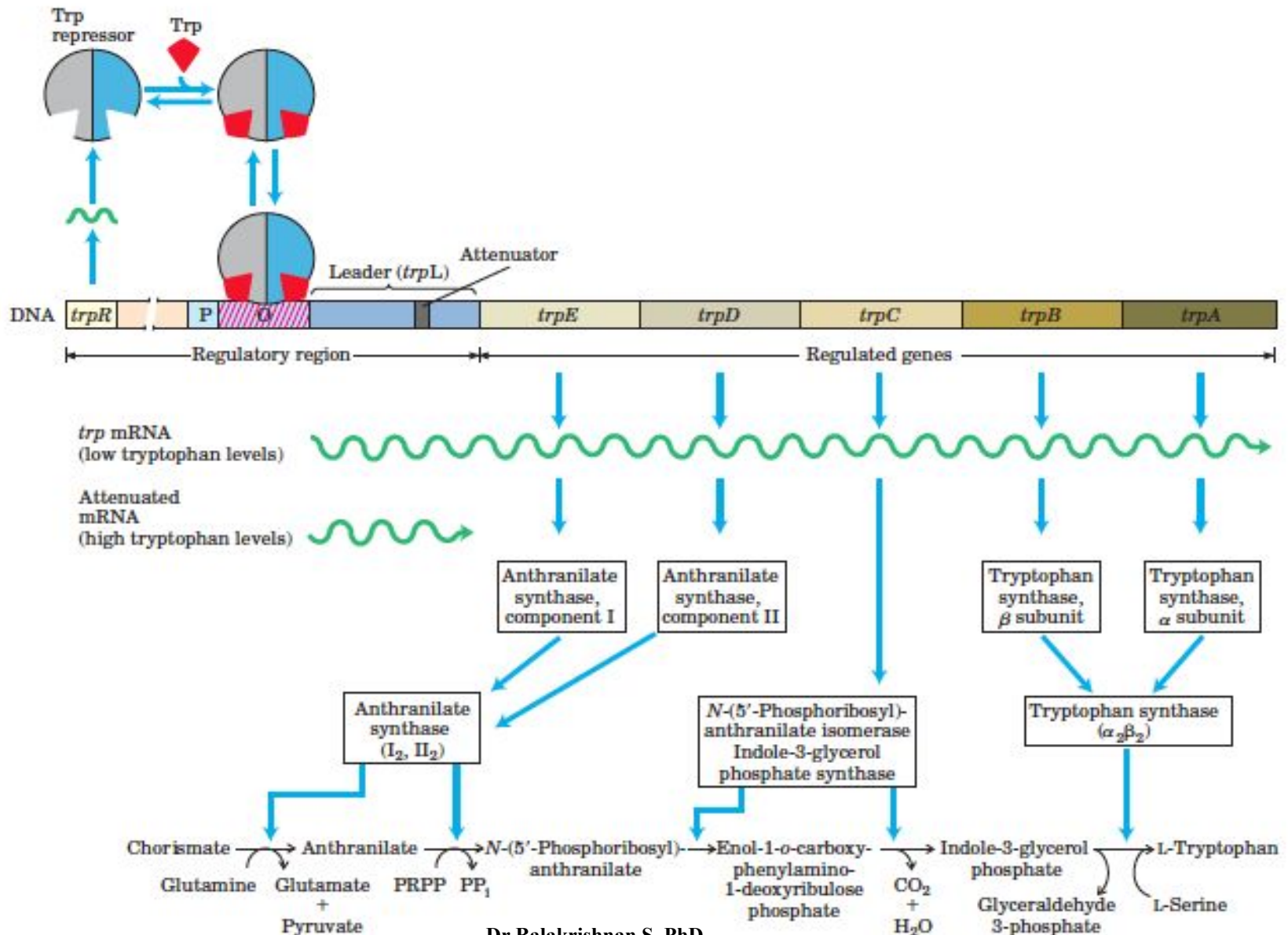
- **Catabolite Activator Protein (CAP) or cAMP Receptor Protein (CRP) is a dimer and acts as a positive regulator of many catabolic operons like the “Lac operon” in E. coli.**
- **Attachment of RNA polymerase to the promoter site requires the presence of CAP bound to cAMP.**
- **Absence of glucose in the cell activates adenylate cyclase which catalyses the synthesis of cAMP from ATP. The latter binds to CAP to form a CAP-cAMP complex.**
 - **Unlike free CAP, this complex binds to promoter site immediately.**
 - **This stimulates the initiation of transcription of the “Lac operon” structural genes in the absence of repressors.**
 - **The binding of CAP-cAMP complex to promoter enhances RNA polymerase activity.**
- **On the other hand, presence of glucose lowers the intracellular cAMP.**
 - **Due to decreased levels of cAMP, the formation of “CAP-cAMP complex” is lowered.**
 - **Hence the binding of RNA polymerase to P-site is negligible and the transcription does not occur.**



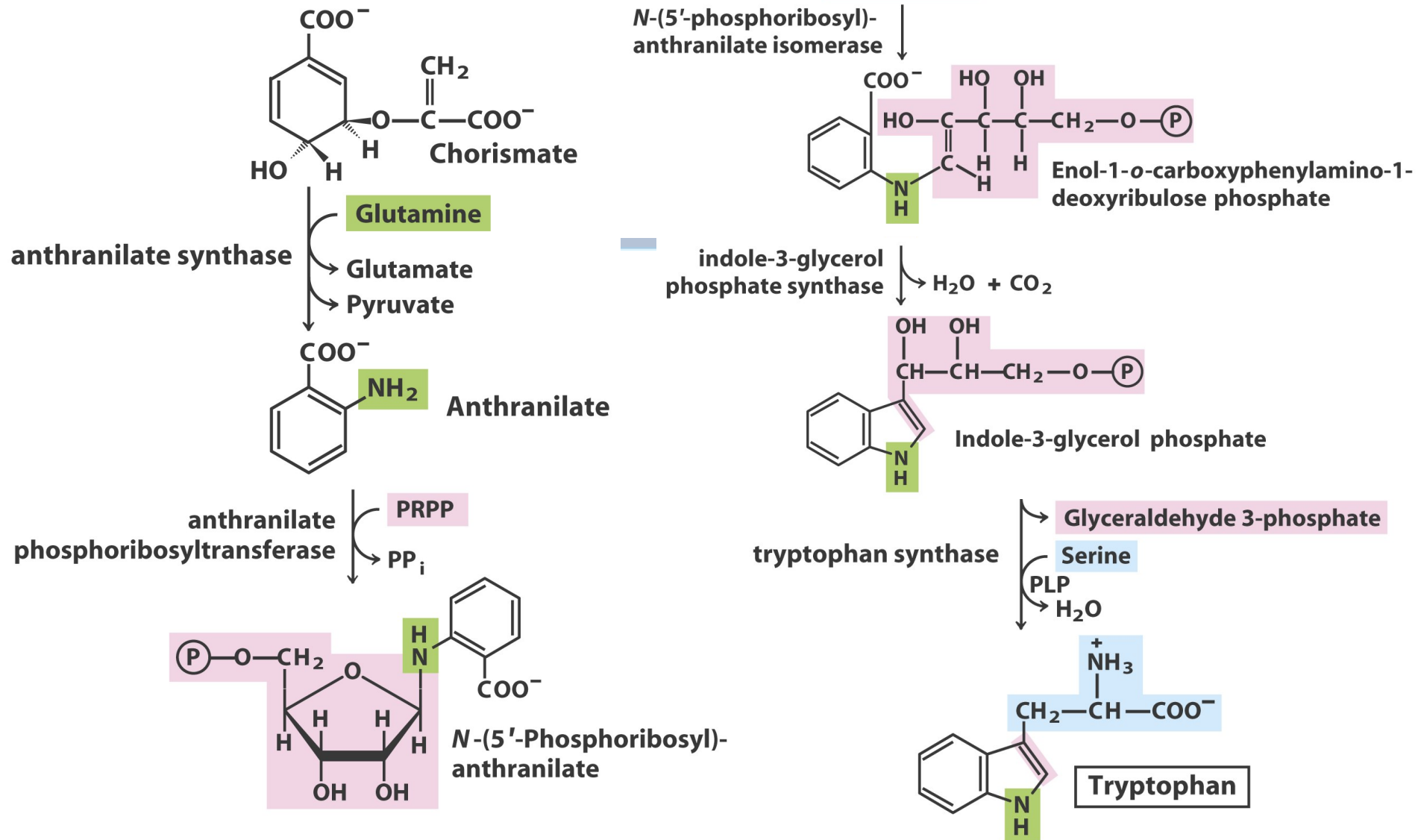
- High levels of transcription take place only when glucose concentrations are low (so cAMP levels are high and CRP-cAMP is bound) and lactose concentrations are high (so the Lac repressor is not bound).
- Without bound activator (CRP-cAMP), the lac promoter is poorly transcribed even when lactose concentrations are high and the Lac repressor is not bound.

Trp Operon:

Many Genes for Amino Acid Biosynthetic Enzymes Are Regulated by Transcription Attenuation – e.g., Trp operon



Synthesis of Tryptophan



Gene Expression in Eukaryotes:

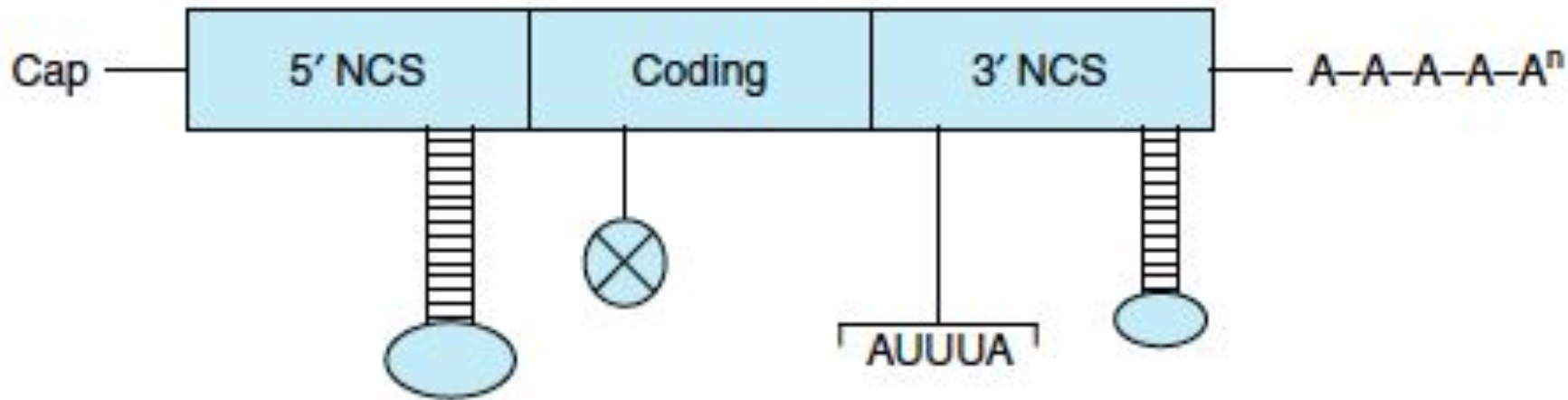
- **Gene expression in the eukaryotes, cannot be the same as that in the prokaryotes because of the following:**
 - **Larger genome**
 - **Redundancy in the DNA**
 - **Repressor proteins influence gene expression are synthesized outside the nucleus and cannot get into the nucleus through the nuclear membrane.**

Methods of Regulation of Gene Expression in Eukaryotes

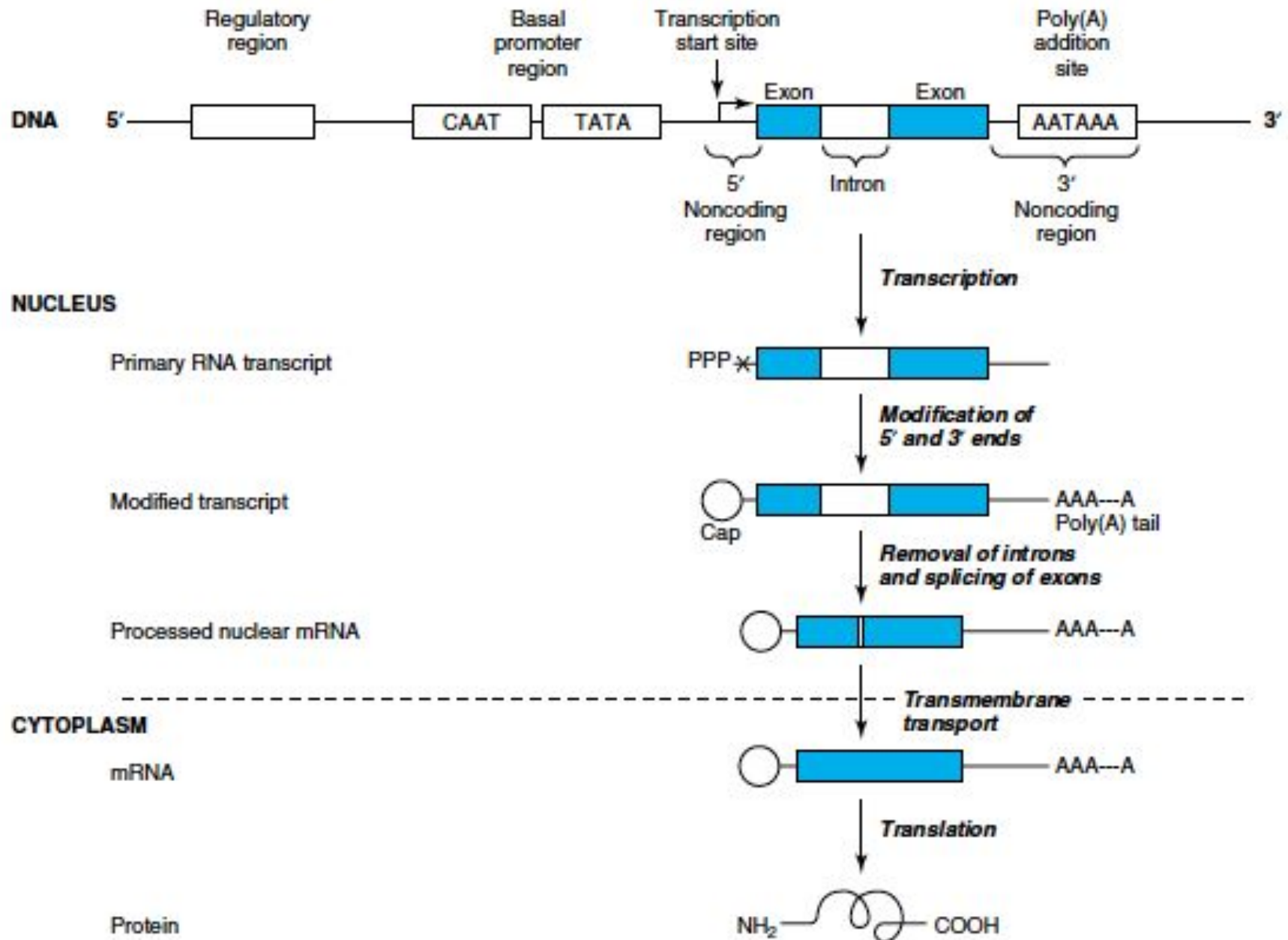
- **RNA processing**
- **Gene amplification**
- **Gene rearrangement**
- **Gene regulation by histones and nonhistone proteins**
- **Class switching**
- **Stability of m-RNA**
- **Binding of regulatory proteins to DNA**
- **Role of enhancers and silencers**
- **Locus control regions (LCRs) and insulators**
- **RNA interference (RNAi)**

RNA processing:

- “Capping” of the 5'-end of the primary transcript
- Addition of a polyadenylate “tail” to the 3'-end of transcription
- And excision of “intron” regions to generate spliced “exons” in the mature m-RNA molecule.

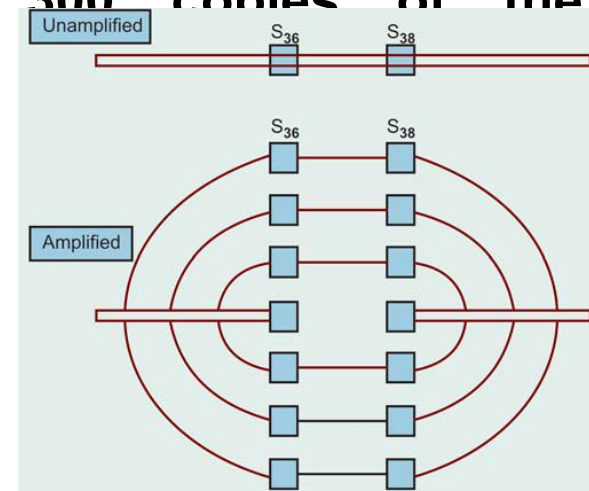


RNA processing:

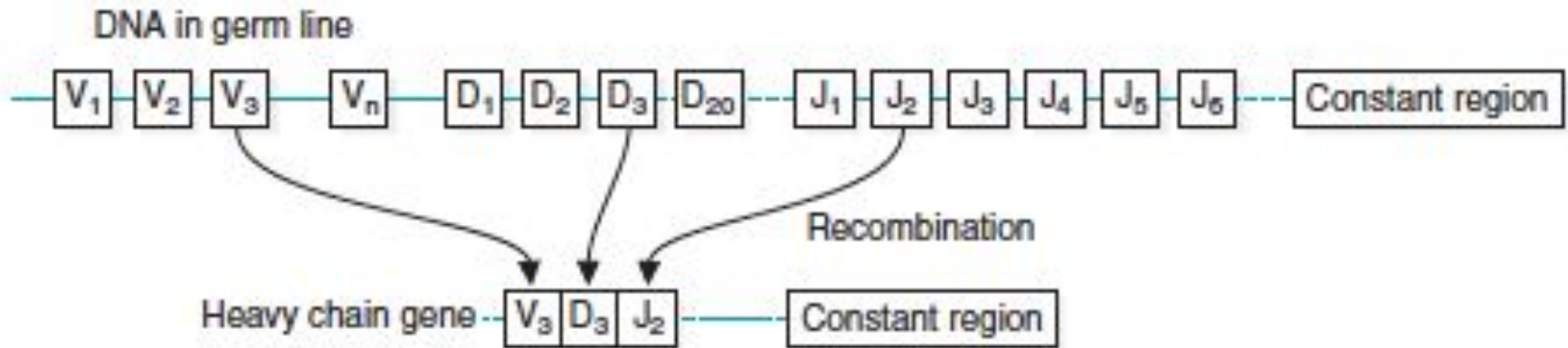


Gene amplification

- In this mechanism, the expression of a gene is increased several-fold.
- This is commonly observed during the developmental stages of eukaryotic organisms.
 - E.g., Amplification of genes for chorion egg shell proteins S36 and S38 during oogenesis in *Drosophila* (Figure below)
- Gene amplification is also common in tumor cells.
 - For instance, in one of the most common childhood solid tumors, neuroblastoma, patients have up to 300 copies of the growth-promoting N-myc gene.



Gene rearrangement



Rearrangement of DNA:

- Both T and B lymphocytes undergo DNA rearrangements, with loss of intervening DNA.
- Such rearrangements create a single transcriptional unit for the formation of a diverse repertoire of antibodies and T-cell receptors for a competent immune system.
- Specific V, D, and J segments from a large number of potential sequences in the DNA of precursor cells combine to form the heavy-chain gene from which lymphocytes produce immunoglobulins (antibodies).

Gene regulation by histones and nonhistone proteins:

- **Histones of nucleosomes can also regulate gene expression.**
 - **It has been found that during S-phase of cell cycle there are many post-translational modifications of the different histones.**
 - ✧ **Include acetylation, methylation, phosphorylation, ubiquitination, sumoylation, citrullination, polyADPriboseylation and N-glycosylation.**
 - ✧ **Such modified histones can regulate gene expression.**

Class switching:

- In this process, one gene is switched off and a closely related gene takes up the function.
- Examples: Class switching is best illustrated by Hb synthesis.
 - In intrauterine life, first embryonic Hb is produced having two 'zeta' and two 'eta' chains.
 - By the sixth month of intrauterine life, embryonic Hb is replaced by Hb-F consisting of $\alpha_2 \gamma_2$ chains.
 - After birth Hb-F is replaced by adult types of HbA, 97 per cent HbA, ($\alpha_2 \beta_2$) and 3 per cent Hb A2 ($\alpha_2 \delta_2$).
- Thus the gene expression is shifted from:

Zeta-eta $\rightarrow \alpha \gamma \rightarrow$ then $\alpha \beta$ and $\alpha \delta$

Stability of m-RNA:

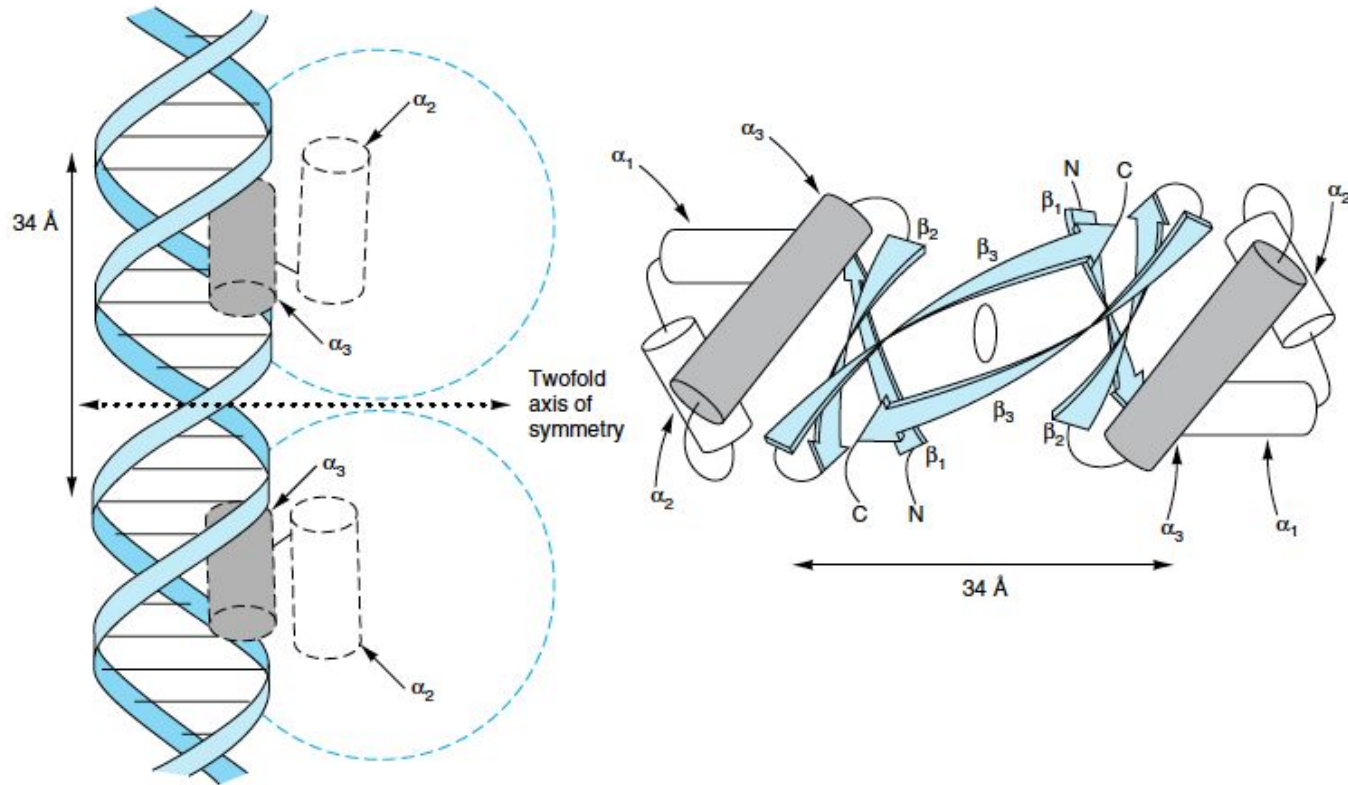
- The stability of m-RNA molecules in the cytoplasm can clearly affect the level of gene expression in a positive or negative direction.
 - There is a potential control site that can be influenced by hormones and other effectors.
 - Some hormones influence the synthesis and degradation of specific m-RNAs.
 - For example, estradiol prolongs the half-life of vitellogenin m-RNA from a few hours to more than 200 hours.
- This coupled with the fact that estrogens enhance the rate of transcription of this gene by four to six-fold results in a tremendous increase of vitellogenin m-RNA.

Binding of regulatory proteins to DNA:

- The regulatory proteins bind with high affinity to the correct region of DNA and control the transcription of certain genes.
- Four unique motifs – account for many of these specific protein - DNA interactions; they are highly conserved between species.
- They are:
 - Helix-turn-helix motif
 - Helix-loop-helix motif
 - Zinc finger motif, and
 - Leucine zipper motif

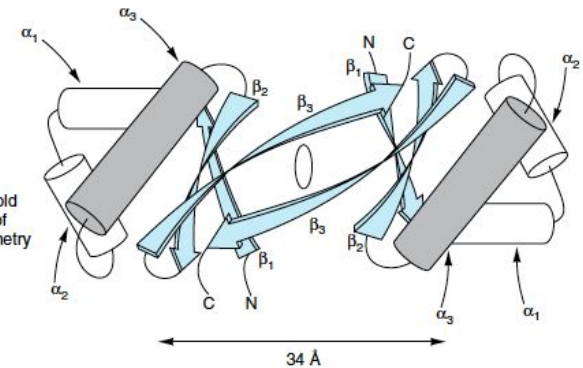
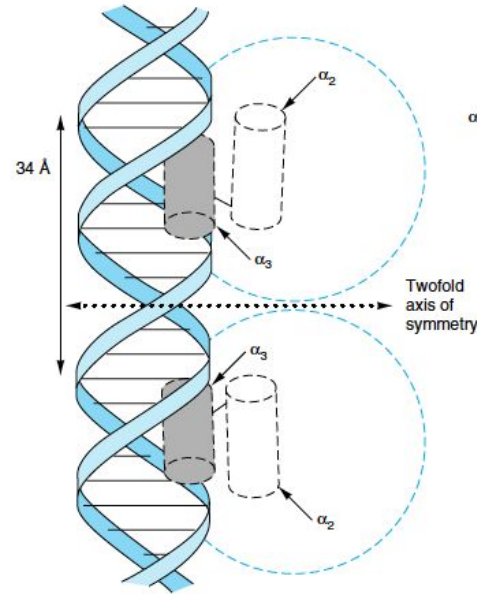
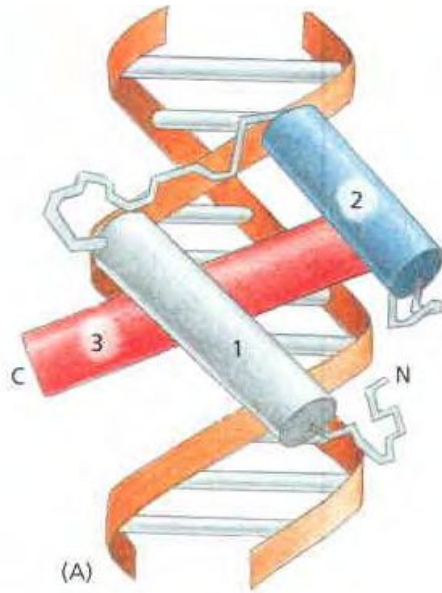
Binding Motif	Organism	Regulatory Protein
Helix-turn-helix	<i>E. coli</i>	lac repressor, CAP
	Phage	λ cl, cro, and 434 repressors
	Mammals	Homeobox proteins Pit-1, Oct1, Oct2
Zinc finger	<i>E. coli</i>	Gene 32 protein
	Yeast	Gal4
	<i>Drosophila</i>	Serendipity, hunchback
	Xenopus	TFIIIA
	Mammals	Steroid receptor family, Sp1
Leucine zipper	Yeast	GCN4
	Mammals	C/EBP, fos, Jun, Fra-1, CRE binding protein (CREB), c-myc, n-myc, l-myc

Helix-Turn-Helix/ Helix-loop-helix Motif:



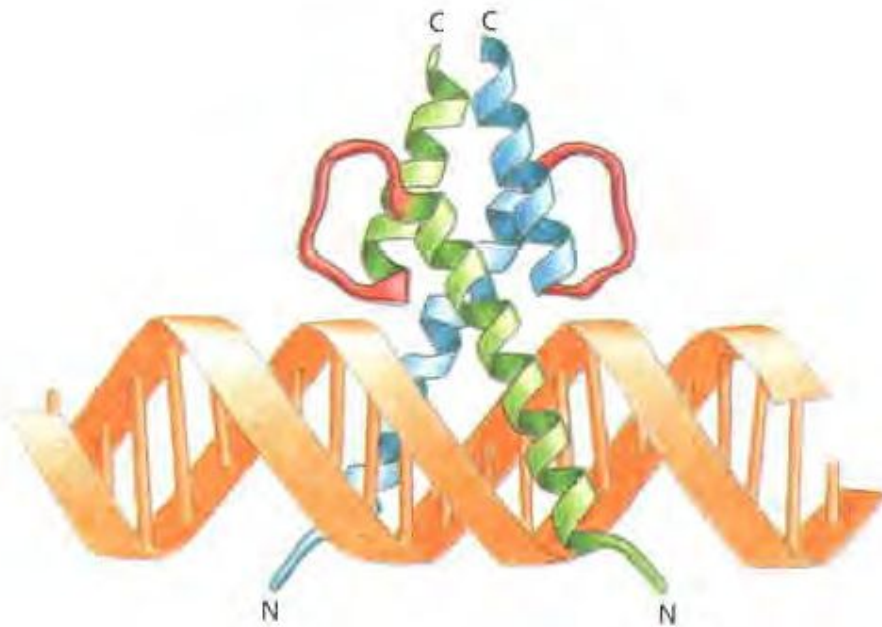
- **Helix-Turn-Helix Motif:**
 - Mediates DNA binding by fitting into the major groove of the DNA helix.
- **Helix-loop-helix Motif:**
 - Promotes both DNA binding and protein-dimer formation.

Helix-turn-helix motif:



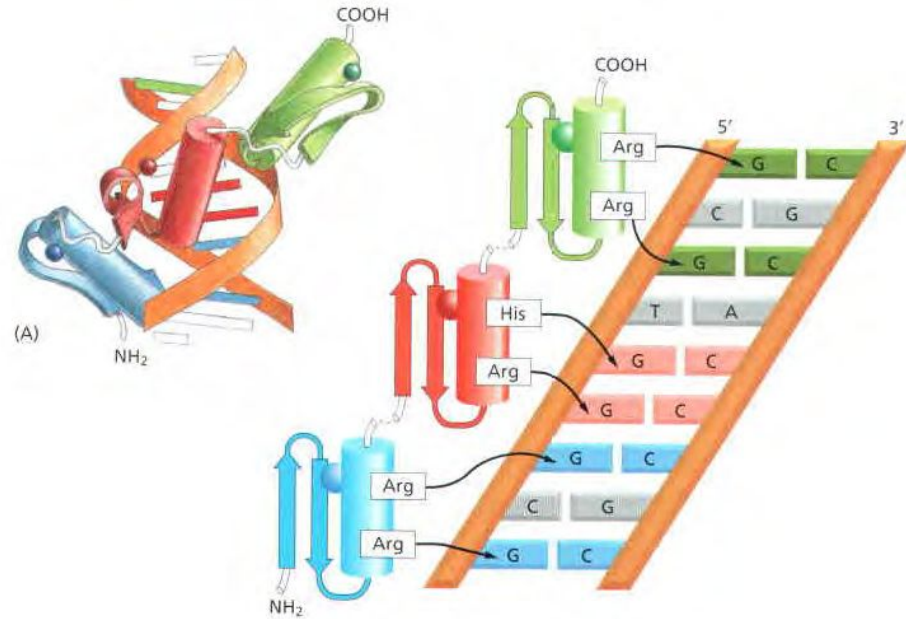
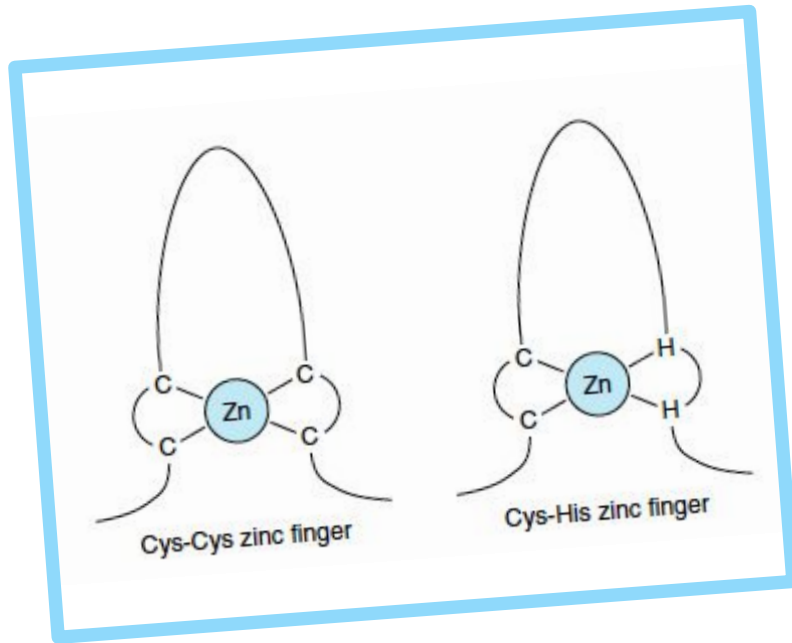
- **Helix-Turn-Helix Motif:**
 - Mediates DNA binding by fitting into the major groove of the DNA helix.
- The homeodomain is folded into three α -helices which are packed tightly together by hydrophobic interactions. The part containing helices 2 and 3 closely resembles the helix-turn-helix motif.

Helix-loop-helix motif



- **Helix-loop-helix Motif:**
 - Promotes both DNA binding and protein-dimer formation.
- The two monomers are held together in a four helix bundle: each monomer contributes two α helices connected by a flexible loop of protein (red). A specific DNA sequence is bound by the two α helices that project from the four-helix bundle.

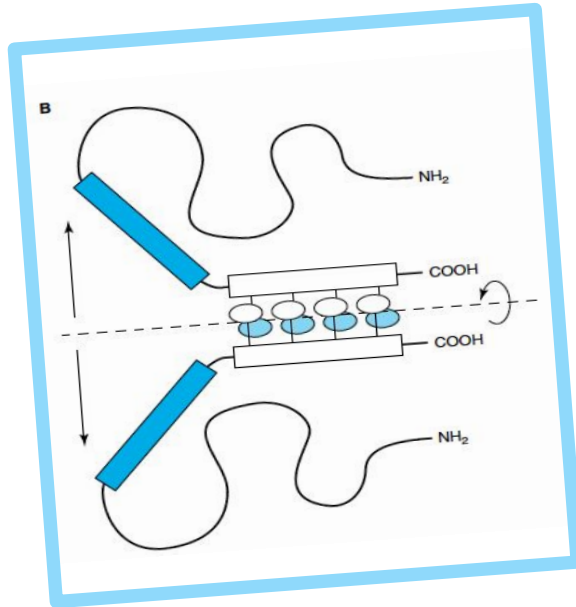
Zinc Finger Motif



- Zinc fingers are a series of repeated domains (two to nine) in which each is centered on a tetrahedral coordination with zinc.
- In the case of TFIIIA, the coordination is provided by a pair of cysteine residues (C) separated by 12–13 amino acids from a pair of histidine (H) residues. In other zinc finger proteins, the second pair also consists of C residues.
- Zinc fingers bind in the major groove, with adjacent fingers making contact with 5 bp along the same face of the helix.
- Proteins of the steroid/thyroid hormone family of transcription factors also contain zinc fingers. (All steroid hormone receptors are zinc finger transcription factors)

Leucine zipper motif

- Leucine zipper forms a dimer that ‘grips’ the DNA helix like a peg, by inserting into the major groove



A schematic model of the DNA binding domain of C/EBP (CCAAT-enhancer binding proteins; transcription factors):

- Two identical C/EBP polypeptide chains are held in dimer formation by the leucine zipper domain of each polypeptide (denoted by the rectangles and attached ovals).
- This association is apparently required to hold the DNA binding domains of each polypeptide (the shaded rectangles) in the proper conformation for DNA binding.

Role of enhancers and silencers:

- **Enhancers:**
 - **DNA segments that promote the synthesis of RNA.**
- **Silencers:**
 - **DNA segments that decrease the transcription.**
- **A number of enhancers, silencers, tissue-specific elements, drug-response elements, hormones/metals have been identified.**
- **The activity of a gene at any moment reflects the interaction of these cis-acting DNA elements with their respective trans-active factors.**

Locus control regions (LCRs) and insulators

LCRs:

- **Some regions of transcription domains are controlled by complex DNA elements called locus control regions (LCRs).**
- **An LCR with associated bound proteins controls the expression of a cluster of genes.**
- **The best defined LCR has been found to regulate expression of the globin gene family over a large region of DNA.**

Insulators:

- **These are DNA elements, in association with one or more proteins prevent an enhancer from acting on a promoter on the other side of transcription domain.**

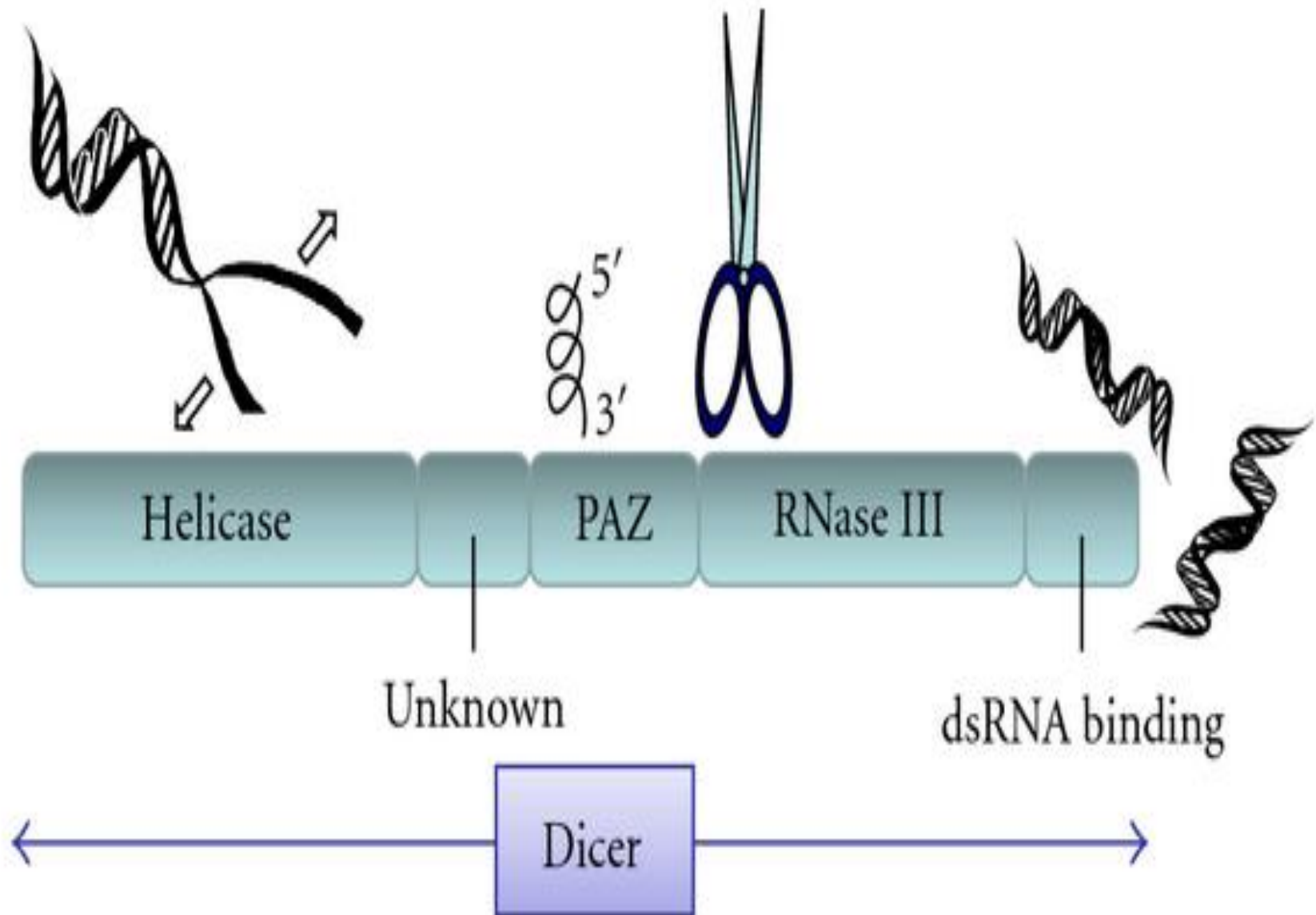
RNA interference (RNAi)

- ◉ **A precise process of mRNA degradation to silence expression of a specific gene.**
- ◉ **Emerging as a new strategy to selectively shut off the post-transcriptional expression of mRNA.**
- ◉ **RNAi is mediated by a short 21– to 25–base pair, double-stranded RNA (dsRNA) - recognizing an exact sequence - functioning as a silencing complex.**
- ◉ **A powerful tool in antiviral research**
- ◉ **Eco friendly, bio safe and evergreen technology**
- ◉ **RNA silencing has roles in regulation of;**
 - **Cell growth,**
 - **Death,**
 - **Differentiation, and**

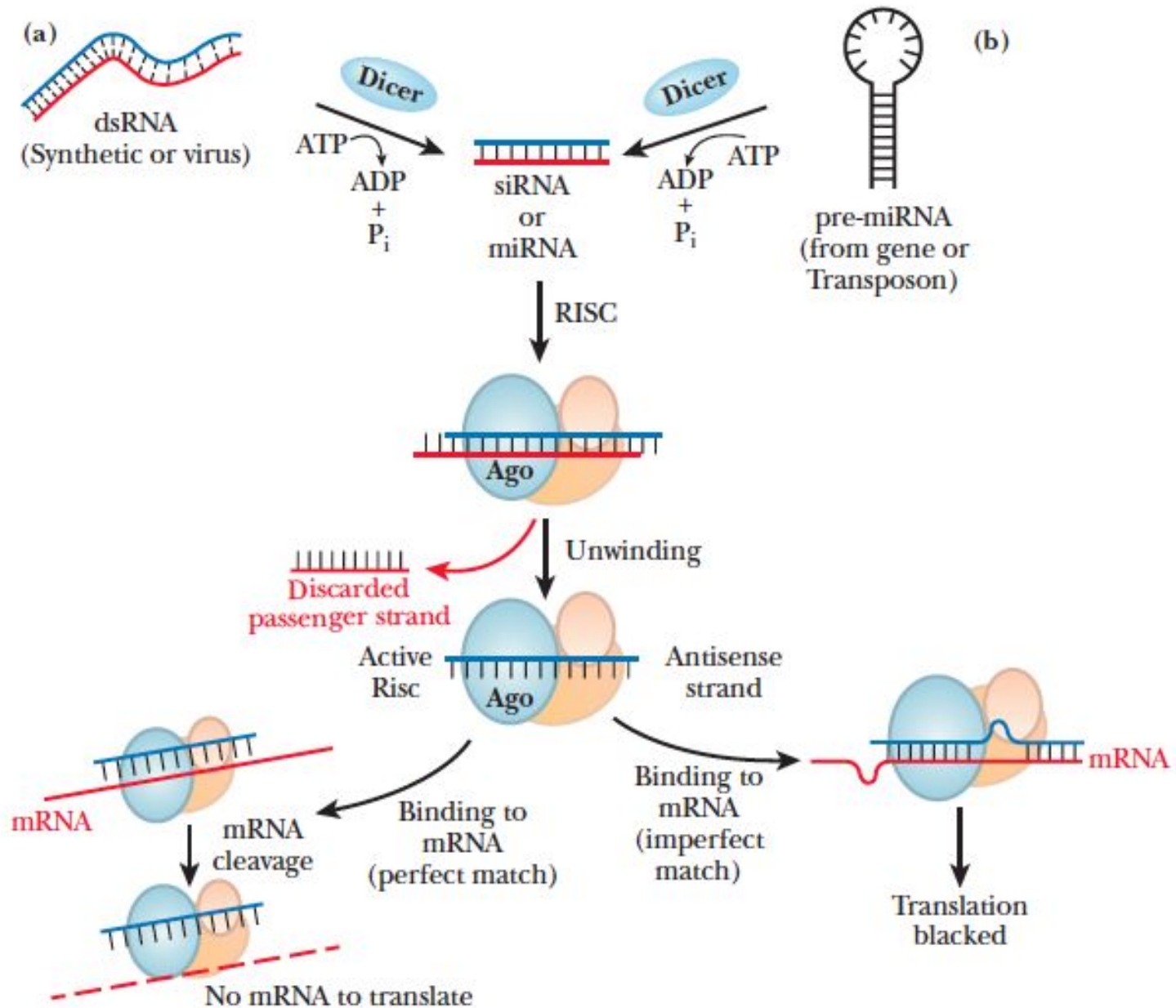
Mechanism of RNA interference:

- **Dicer:**
 - An enzyme, which process the introduced or synthesized double stranded RNA (dsRNA) inside the mammalian cells.
 - This results in formation of dsRNAs, about 21 nucleotides in length—the small interfering RNAs (siRNAs)/ miRNA.
- siRNA/ miRNA coupled with cellular proteins forms RNA interfering silencing complex (RISC) and heads for target recognition on intracellular mRNAs.
 - RISC of siRNA binds to complementary sequences on mRNA and effectively leads to the cleavage of mRNA.
 - RISC of miRNA is not a perfect match for the mRNA target, so there is no cleavage of the mRNA, but the RISC sticks to the mRNA, preventing its translation.
- The same can also be expressed intracellularly when introduced through plasmid or viral vectors.
 - Synthetic siRNA can be easily introduced into mammalian cells using transfecting agents or by electroporation

DICER PROTEINS:



Mechanism of RNA interference:



Read more....



Thank you...